E4-20
E9AC3

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2013

MSc and EEE PART IV: MEng and ACGI

FULL CUSTOM INTEGRATED CIRCUIT DESIGN (SOLUTIONS)

Friday, 10 May 2013 10:00am

Time allowed: 2 hours

There are FOUR questions on this paper.

Answer THREE questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

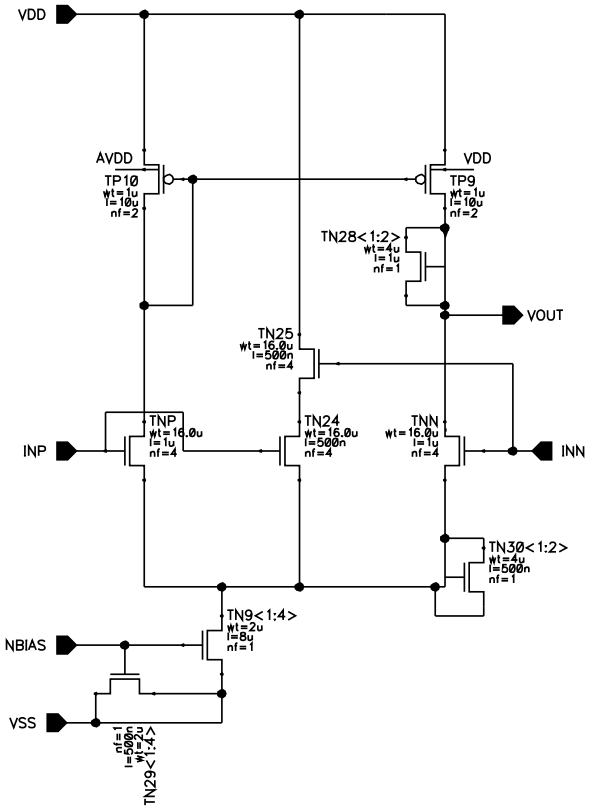
Examiners responsible — First Marker(s): T.G. Constandinou
Second Marker(s): C. Mehring

Solutions annotated as follows:

Black – model answers
Red – marking scheme
Blue – examiners feedback after marking

Question 1

(a)



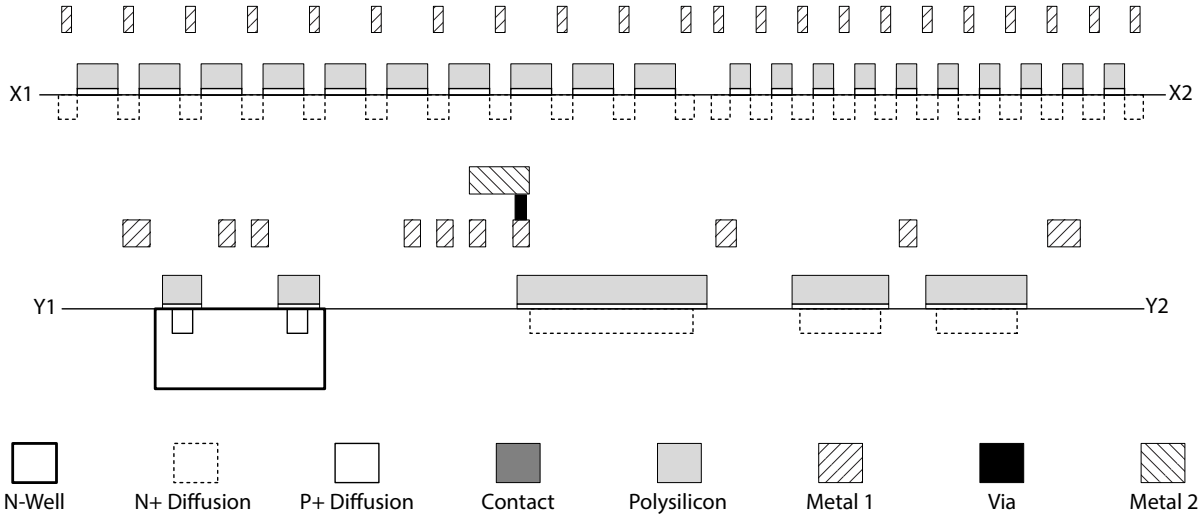
A=INP, B=INN, C=VOUT, D=NBIAS and device M1=TN28<1>

(4 marks for identifying device W/L's correctly, 1 mark for identifying dummy devices, 4 marks for correct netlist, 1 mark for correct pin identification)

[10 marks]

Most answers failed to combine fingered (i.e. parallel connected) devices into single devices in schematic. Also dummy devices generally were not identifies as such.

(b)



[8 marks]

Generally answered very well.

(c) This circuit is a single stage OTA (operational transconductance amplifier) with bump linearlisation (devices TN24 and TN25). By "stealing" current from the differential pair the middle branch the (linear) input voltage range is increased.

(1 mark for identifying OTA topology, 1 mark for bump linearization)

[2 marks]

Very few students identified this was an amplifier- most answers thought this was some type of logic. Generally, the give-away clue that this was an analogue (and not digital) circuit is that device sizes were not minimum length.

Question 2

(a) Any 6 of the following with corresponding cross-sections (standard):

- MOSFET (NMOS and PMOS) – native devices so good performance.
- Lateral PNP bipolar transistor (p-substrate/n-well/p+) – limitation is gain (beta). Use is limited to bandgap reference and diode.
- Poly1-poly2 capacitor – typically 0.5-1fF/sq. um
- Parasitic p-n junction diodes (n-well/p-sub, p+/n-well, n+/p-sub) – can be used as photodiodes.
- Poly resistor – high value resistors hard to realize (even if technology supports a high resistance poly).
- Lateral (flux) interconnect capacitors. Generally capacitance is less than poly1-poly2.
- Photodiode (n+/p-epi/p-substrate).
- ISFET – leaving floating gate connected up to M3 and using passivation surface as sensing membrane.

(For each: ½ mark for identifying topology + ½ for description + ½ for limitations + ½ for cross-section)

[12 marks]

Generally answered well, however many students did not illustrate with side-views.

(b)

- Generally pixel size has scaled with Moore's law if you look at Mpixels/mm^2 over the past decade or so.

[1 mark]

Very few answers made reference to Moore's law.

- Components within 3T pixels are typically photodiode (or photogate) and 3x Transistors. Transistors scale linearly with technology, and therefore so do dimensions of structures used to design photodiode (or photogate). However, dark current gets worse (with scaling) so good performance imagers are harder to make in smaller technologies.

[1.5 marks]

Not addressed very well in general.

- Imaging plane (i.e. pixel array) would scale from 350nm to 32nm with Moores law (i.e. in a 32nm technology pixel size could be expected to be <1um x 1um. However- auxillary circuits are the fundamental bottleneck to scaling. For example a 200Mpixel imager (at 5 frames per second) and 10 bit resolution data converter corresponds to a datarate of 10Gbps! So as the imaging array scales the requirements on the FSM, column amplifiers, array amplifier, data converter, etc increase.

[1.5 marks]

Answers did not look from system-level (in general). Mostly concentrated just on pixel-size and not amplifiers, ADC, etc- the data throughput problem.

(c)
Using fact that shorter wavelength light is absorbed at shallower depths means that longer wavelength light penetrates deeper into the silicon (before being absorbed). If the designer can exploit the depth of different type of junctions (for example, wells are typically at 1-2 micron depth, whereas diffusion is shallow – 100-200nm) such as to separate an electron-hole pair and create a photocurrent, then different depth junctions can be used to detect different wavelengths of light. Specifically- diffusion should be sensitive to all wavelengths, whereas shallow wells only to red+green and deep wells only red. Foveon (X3 imagers) actually exploit this phenomenon to achieve vertically stacked pixels (i.e. detect R + G + B within every pixel) as oppose to using colour filters in a Bayer pattern and then interpolating the "missing pixels".

(2 marks for general understanding and identifying that different structures sensitive to different wavelengths, 1 mark for specific examples (and giving numbers), 1 mark for example with Foveon, etc)

[4 marks]

Answered very well.

Question 3

(a)
- If a 1V output is required given 10K resistance this corresponds to a 100uA current (full scale).
- Therefore if I(LSB) x (2^N-1) = 100uA then I(LSB)=0.4uA.
- Given IREF is 2uA then the ratio (MREFA:M1A:M2A:M3A:...M8A and MREFB:M1B:M2B:M3B:...M8B) should be: 5:1:2:4:8:16:32:64:128

(2 marks for method + 2 marks for answer)

[4 marks]

Generally answered well.

(b) Design question so no unique answer. Looking for the following:

- For first 4 LSBs need ratio 5:1:2:4:8 giving a total of 21 LSB devices. Could arrange in a 3x7 array or alternatively 5x9 array (with dummy devices around). Large devices should be made from identical unit devices.
- Use of either common centroid or interdigitation (1D or 2D)
- Currents should flow in same direction (and devices orientated similarly).

(2 marks for demonstrating good layout practice + 2 marks for correct answer)

[4 marks]

Generally answered well – some students tried to draw full layout! (not needed).

(c)
Advantages of current-steering over R-2R

- Current output (if application requires a current) is MOSFET only.
- In general (for same area) MOSFETs match better than Rs or Cs.
- Using currents, can be much lower power than resistors (since large resistors cannot be integrated).

Advantages of R-2R over current steering.

- Not binary weighted- therefore less total unit components (3R x (N+1)) vs. 2^N-1.
- Easier to match since all passives are either R or 2R.
- Voltage output does not depend on absolute resistance value (as in case of current-steering).

(4 marks for 4/6 items from above)

[4 marks]

Generally answered well – advantages/disadvantages relating to power + area were incorrect for many answers.

(d)
$V_{GS}=V_{TH}+\sqrt{(2I_D/\mu C_{ox}W/L)}$

1. If $(W/L)M_{REF}$ = 25/25, $V_{GS}=0.5+\sqrt{(2*2\mu A/80\mu)}$ = 0.7236V.
2. Since ratio of M_{REF}: M_{LSB}=5:1 then $(W/L)_{LSB}$ = 5/25.
3. $\sigma(\Delta V_{TH})=AV_{TH}/\sqrt{(WL)}$ = 4.5mV/√(20) = 0.4mV => $3\sigma(\Delta V_{TH})$=1.21mV (worst case).
4. $\Delta I_D(LSB, max)=|I_{DLSB}(V_{GS}-V_{TH})-I_{DLSB}(V_{GS}-V_{TH}-\Delta V_{TH})|=|0.4\mu A-0.389\mu A|$ =4.31nA.
5. For mismatch not to effect DAC, 2^N-1 x $\Delta I_D(LSB, max)<0.5*I_{LSB}$
6. 2^N-1 x $\Delta I_D(LSB, max)$ = 1.1µA > $0.5*I_{LSB}$!!!
7. ENOB= $\log2(0.5xI_{LSB}/\Delta I_D(LSB, max))$ = log2(200nA/4.31nA)=4.54bits

(4 marks for method, 1 mark for 3sigma worst case, 3 marks for ENOB)

[8 marks]

No students attempted this answer!

Question 4

(a)

$$V_{GS1} = V_{GS2} + I_{out}R_S$$

$$\sqrt{\frac{2I_{out}}{\mu_n C_{ox}(W/L)_N}} + V_{TH1} = \sqrt{\frac{2I_{out}}{\mu_n C_{ox}K(W/L)_N}} + V_{TH2} + I_{out}R_S$$

$$\sqrt{\frac{2I_{out}}{\mu_n C_{ox}(W/L)_N}}\left(1-\frac{1}{\sqrt{K}}\right) = I_{out}R_S$$

$$I_{out} = \frac{2}{\mu_n C_{ox}(W/L)_N}\frac{1}{R_S^2}\left(1-\frac{1}{\sqrt{K}}\right)^2$$

(3 marks for Iout derivation, 2 marks for expression, 1 mark for it being independent of VDD)

[6 marks]

Correctly answered in general.

(b)

$Iout=(1/800K)^2*(1-0.5)^2*2/(200u)$=3.9nA.

(1 mark for correctly identifying K and 1 mark for result)

[2 marks]

Correctly answered in general.

(c)

$\beta_{14}/\beta_{16}=(W_{14}L_{16}/L_{14}W_{16})=[(V_{DD}-V_{SPH})/(V_{SPH}-V_{THN})]^2=4.5/14.5=0.31 \Rightarrow \sqrt{(\beta_{14}/\beta_{16})}=0.56$

$\beta_{11}/\beta_{15}=(W_{11}L_{15}/L_{11}W_{15})=[(V_{SPL})/(V_{DD}-V_{SPL}-V_{THP})]^2=\beta_{11}/\beta_{15}=4.5/11.5=0.39 \Rightarrow \sqrt{(\beta_{11}/\beta_{15})}=0.62$

$0.56=(1.8-VSPH)/(VSPH-0.4) \Rightarrow VSPH=1.3V$

$0.62=(VSPL/(1.8-VSPL-0.5) \Rightarrow VSPL=0.5V$

(2 marks for each correct expression + 1 mark for evaluating each)

[6 marks]

Correctly answered in general.

(d)

$t1=C(VSPH-VSPL)/I=50p(0.8)/3.9n=10.26ms$

$t2=C(VSPH-VSPL)/I=50p(0.8)/3.9n=10.26ms$

$fclk=1/(t1+t2)=48.75Hz$

(1 mark for correct expression + 1 mark for evaluating)

[2 marks]

Correctly answered in general.

(e)

Area=Area(MOSFET) + Area(Cap) + Area (Resistor) + Routing

1. Area(MOSFET)=7*10*10+5*5+3*0.18*2+2*4.5*0.18+11.5*0.18+14.5*0.18=732.4um2. Add 200% overhead for contacts+spacing = 2200um2
2. Area(Cap)=50pF/0.7fF=71428um2
3. Assuming Resistor = (L/W)=800/1 with W=1um. Area(Resistor) = 800um2
4. Add extra 50% for routing. Total area = (2200+71428+800) * 1.5 = 111642um2 = 0.11mm2

(1 mark for cap, 1 mark for resistor, 1 mark for MOSFETs+assumptions, 1 mark for final result)

[4 marks]

Correctly answered in general – however many students did not allow any overhead for wiring/interconnects/contacts.